Bluetooth hardware, operating system support starts at the HCI layer (see Figure 4).

A closer look at the Bluetooth protocol stack

The Bluetooth protocol stack is divided into two parts – the host and controller, and interfaced by the host controller interface (HCI) layer (see Figure 5).

Bluetooth host

The host consists of the software part of the stack. Hence, it is implemented at the operating system level. It includes implementations of the core Bluetooth protocols: the Bluetooth stack (or host protocol stack) and the high-level layers of the Bluetooth architecture, such as APIs and profiles. It has various layers and protocols like L2CAP, RFCOMM, SDP, etc, which are explained below.

L2CAP(Logical Link Control and Adaptation Protocol) L2CAP has the following features:

- The capability to multiplex data between different higher layer protocols; hence, various protocols like RFCOM, SDP, etc, can operate over it
- Segmentation and reassembly
- Quality of service
- It supports a group abstraction, enabling implementation for mapping a group on to a piconet

RFCOMM (Radio Frequency Communication): RFCOMM is situated over the L2CAP layer. It is a serial port emulation protocol. It emulates RS-232 control and data signals over the Bluetooth baseband.

It provides roughly the same service and reliability guarantees as TCP. The choice of port numbers is the biggest difference between TCP and RFCOMM from a network programmer's perspective. Whereas TCP supports up to 65,535 open ports on a single machine, RFCOMM allows only 30. This has a significant impact on how to choose port numbers for server applications.

SDP (Service Discovery Protocol): This protocol is required by all Bluetooth applications and is an important part of the Bluetooth protocol stack. With the help of this protocol, devices can discover what services other nearby devices support and what parameters are required to connect with them. Each service is identified by a universally unique identifier (UUID).

Bluetooth profile

The Bluetooth profile is a specification that describes how devices must use Bluetooth protocols to implement a particular task, and this is the theoretical part of the Bluetooth architecture. Bluetooth services are the practical part of the Bluetooth protocol stack. There are various types of profiles available. Some are:

- ***GAP (generic access profile):*** This defines generic requirements for detecting and establishing a connection to a Bluetooth device. It is a core profile that is available by default.
- ***GOEP (generic object exchange profile):*** This defines the requirement of OBEX, a communication protocol used to exchange binary objects between devices while using minimal resources and usage models.
- ***HSP (headset profile):*** This defines the procedure that supports the interoperability between a headset and a mobile device, like a cellular phone, that the headset controls through AT commands, and depends upon the serial port profile.
- ***SPP (serial port profile):*** This defines the requirements for virtual serial port connections over the RFCOMM.

Note: To see the available Bluetooth profiles in your Linux system, run *sdptool browse local*.

HCI (Host Controller Interface)

The Bluetooth host and Bluetooth controller communicate with the help of the HCI. It contains drivers that abstract and transfer data between the Bluetooth host and the Bluetooth controller. These drivers implement communication between the Bluetooth host and the Bluetooth controller with a small set of functions that send and receive commands, data packets and events.

Communication between the host and the controller is done through HCI packets, of which there are four types.

- ***Command packet*:** These packets are generated from the host and are sent to the controller or Bluetooth adapter to control the adapter. They can be used to start a device enquiry, connect to a remote device, etc.
- ***Event packet*:** These packets are sent from the Bluetooth adapter to the host. These packets are sent whenever an event occurs such as sending information about a local Bluetooth adapter, connecting to a remote device, etc.
- ***ACL data packet*:** ACL data packets are encapsulated by the HCI layer and transport to the Bluetooth adapter; however, the HCI headers are removed and the bare ACL packet is transmitted over the air and re-wrapped after being received.
- ***Synchronous data packet*:** These packets are also encapsulated at the HCI layer and pass to the Bluetooth adapter. Then, after unwrapping the HCI header, they are sent out, but are also re-wrapped after being received.

Bluetooth controller

The Bluetooth controller is a Bluetooth device such as a USB dongle. It is a hardware and firmware part of the Bluetooth protocol stack. It implements the lowest level of Bluetooth architecture, including the physical layer and the radio transceiver. The Bluetooth controller receives commands from the Bluetooth host through the HCI. The Bluetooth controller has layers like the Link Manager Protocol (LMP), baseband and radio.

Link Manager Protocol (LMP): This is responsible for logical link establishment between Bluetooth devices. It does link authentication and encryption, and creates, updates and removes logical links and logical transports. It also updates parameters for a physical link to Bluetooth devices.

Baseband: This layer has a link controller, which is